nTZS x Selcom - Security Controls & Flow Review

Prepared by: NedaPay (NEDA Labs)
Date: 6 July 2026
Re: Requested controls for the unified Selcom custodian / disbursement account (mint + burn), ahead of go-live
Audience: Selcom - Dhimant Jadeja (cc Sameer, Alyoce FM, Rosario Arun)

Context

nTZS is a fully fiat-backed stablecoin: every token in circulation is backed 1:1 by TZS held at Selcom (SMFB) as custodian. In the proposed flow, a single Selcom account is credited on mint (purchase) and debited on burn (redemption).

Consolidating to one account gives us clean 1:1 reconciliation - but it also means the entire reserve is reachable by the disbursement API. The controls below cap the blast radius of any credential compromise while preserving that clean reconciliation. Most are requests to confirm/enable on Selcom's side; a few are questions.

1. Account-level controls (Selcom)

Because a single signing-key compromise could otherwise reach the full reserve, we would like to enforce:

#	Control	Purpose	Proposed starting value (to tune to live volume)
1	Per-transaction cap	Bound any single payout	<= 5,000,000 TZS; above -> manual approval
2	Daily disbursement cap	Bound total daily API payout	~50,000,000 TZS / day
3	Velocity limit	Stop rapid drain	<= 10,000,000 TZS/hour or <= 20 payouts/hour
4	Admin-approval threshold (dual control)	Second-human sign-off on large moves	single > 2,000,000 TZS, or cumulative > 20,000,000 TZS/day
5	Production IP whitelisting	API accepts calls only from our fixed egress IP(s)	Enabled in production
6	Reserve segregation (preferred)	Expose only an operational float to the API; hold the bulk of the reserve in an SMFB sub-account the disbursement API cannot debit	Confirm feasibility

Questions: Which of #1-#4 can Selcom's Transaction Rules / Admin Approvals enforce today? And for #6 **-** can the "unified" account still be structured so the API only ever reaches a float, with the reserve held behind it?

2. Mint / burn trigger endpoints (NedaPay <- Selcom)

Selcom's flow calls our POST /mint and POST /burn. **POST /mint authorizes token issuance, so it must be non-forgeable.** To secure it, we need from Selcom:

- Signed callbacks - an HMAC/RSA signature (or mutual TLS) on the callback so we can verify authenticity. Today the disbursement callback carries no signature.
- A stable idempotency key on each callback (reference_id / selcom_receipt) so retries cannot double-mint or double-burn.
- Static source IP(s) for Selcom's callback origin, so we can allowlist them.

On our side we will: verify the signature + source IP, independently confirm settlement via GET /v1/transaction/query before minting (never mint on the instruction alone), dedupe by reference, and assert on-chain supply == reserve balance after every mint/burn.

3. Frequency & abuse controls (below-threshold attempts)

Per-transaction caps alone don't stop a "salami" drain - many payouts each under the cap. The aggregate limits in Sec. 1 (#2-#4) are the primary defence; to close the gap we additionally ask:

- Count-based velocity limits, not only amount-based - e.g. <= 20 payouts/hour and <= 200 payouts/day via the API, regardless of size. A flood of tiny payouts should be bounded too.
- Beneficiary-level limits - max amount and count per destination account per day (e.g. <= 3 payouts / <= 5,000,000 TZS per beneficiary per day).
- New-beneficiary cooldown - a delay or lower cap for first-time destination accounts, so a compromised key can't immediately drain to fresh mule accounts.
- Limit-trip alerting & auto-suspend - when any rule in Sec. 1 or above trips, notify us in real time (email/webhook) and optionally auto-suspend API disbursements pending manual re-enable.

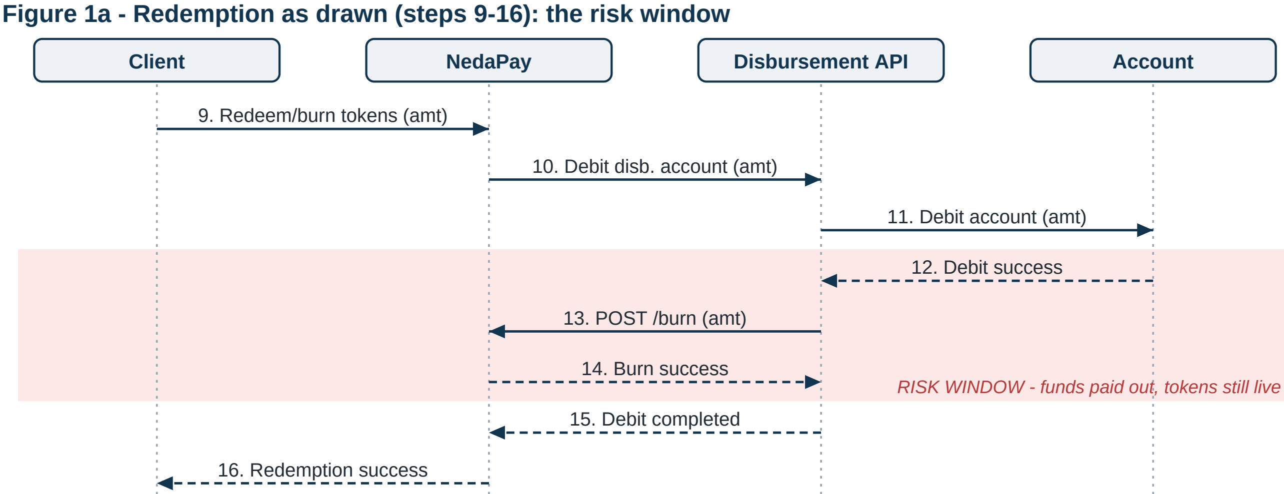
We will mirror equivalent limits in our application layer (rate limiting, per-user redemption velocity, circuit breaker) so neither side is a single point of failure.

Question: which of these can Transaction Rules express today (count-based, per-beneficiary, cooldowns), and does the portal support limit-trip notifications?

4. Two flow adjustments to discuss

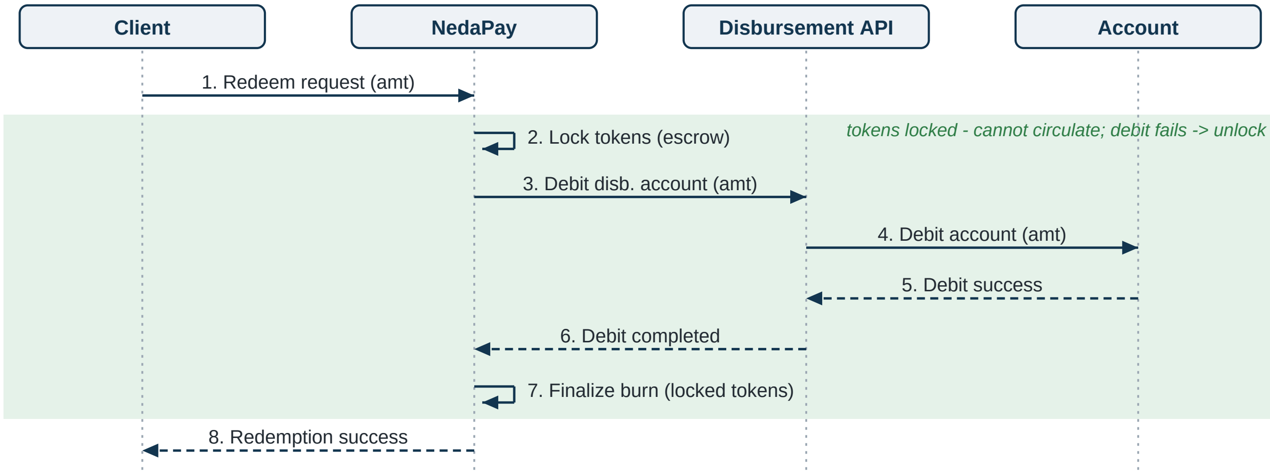
- Burn ordering - the flow debits the account (steps 10-12) before the tokens are burned (13-14). A failure in between leaves funds paid out but tokens un-burned (under-collateralized). Preferred: reserve/lock the tokens first, or burn-then-pay, or a two-phase authorize -> confirm so we can guarantee atomicity. At minimum we will run a reconciliation to catch "debited but not burned".
- Mint entry point - the flow shows NedaPay instructing "Credit Disbursement Account" (step 2), but a real customer deposit is client-initiated (USSD / control number). We will gate minting on the client's actually-received, settled funds. Please confirm how customer collections settle into this account once the collection (push-USSD / control-number) flow is live.

Figure 1a - Redemption as drawn (steps 9-16): the risk window



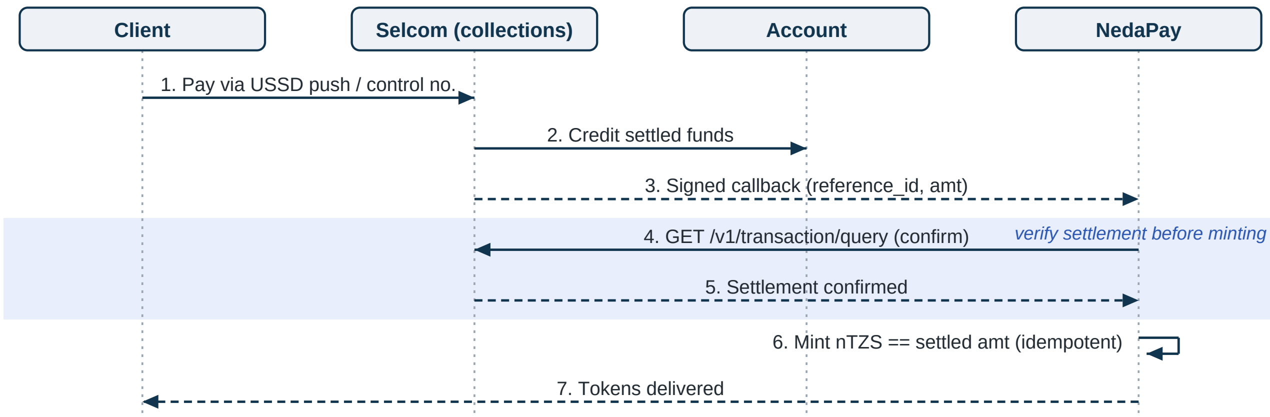
Solid arrows = requests, dashed = responses. Between steps 12 and 14 the money has left the account but the tokens still exist - a failure here leaves nTZS under-collateralized.

Figure 1b - Proposed ordering: lock -> debit -> finalize burn



Tokens are locked before any money moves. If the debit fails they unlock (no loss); if anything fails after the debit, the locked tokens cannot circulate and reconciliation completes the burn.

Figure 2 - Mint entry point: gate on settled funds



Minting is triggered by the client's actually-settled deposit and independently confirmed via the transaction-query API - never by an unverified instruction or callback alone.

5. Open items to close on the call

- Confirm which controls in Sec. 1 are available now vs. need enablement, and Sec. 1.6 feasibility.
- Signed callbacks + static callback IP(s) for Sec. 2.
- Count-based / per-beneficiary rules + limit-trip notifications (Sec. 3).
- Collections (push-USSD / control-number): ETA, settlement timing (T+0 / T+1), and how funds land in this account.
- Production FI codes (Vodacom / bank) + valid purpose codes.
- Government levy applicability on wallet/bank sends (charges are inclusive of VAT + excise; levy applies "where applicable").

Proposed limit values are conservative starting points and will be tuned to observed transaction volumes after go-live. Our disbursement integration (wallet, bank, and internal rails) has already been validated end-to-end against the Selcom sandbox.